01 - Project Overview

FinTech Cybersecurity and Consumer Trust

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

README.md; documentation/project_overview.md; config/settings.yaml; docker-compose.yml

What the project is

This is a final-year research and development project. It builds a Python-based research platform (no web front end) that studies how cybersecurity capability in FinTech - fraud detection, anomaly detection, threat monitoring - relates to consumer trust, and that exposes the live state of the experiment through Prometheus, Grafana and Loki.

One-paragraph summary

Synthetic financial transactions, synthetic security events and a synthetic consumer survey are generated with a fixed random seed. A pipeline validates and cleans the data, engineers features, trains and tunes three supervised models (Logistic Regression, Random Forest, XGBoost) and two unsupervised detectors (Isolation Forest, Local Outlier Factor), evaluates them on a chronological test set, classifies risk into tiers, scores consumer trust from Likert answers, stores results in PostgreSQL and publishes metrics. Docker Compose starts everything with one command.

The big idea in simple words

A bank app can be technically very secure but customers may still not trust it; or customers may trust it while it is weakly protected. The project measures both sides - how well the system catches fraud (measured) and how safe people feel (surveyed) - and watches the whole experiment on dashboards.

Main components

Component	Where	Status
Research pipeline (collect to report)	src/pipeline.py + src/* packages	IMPLEMENTED
Fraud + anomaly models	src/fraud_detection, src/anomaly_detection	IMPLEMENTED
Consumer trust scoring + statistics	src/trust_scoring, src/model_evaluation	IMPLEMENTED
PostgreSQL schema and persistence	database/, src/database/db.py	IMPLEMENTED
Prometheus exporter and live replay worker	src/metrics/exporter.py, src/worker.py	IMPLEMENTED
Grafana dashboards, datasources, alerts	dashboard/grafana, dashboard/prometheus	IMPLEMENTED
Docker Compose stack (8 services)	docker-compose.yml, infrastructure/	IMPLEMENTED
16 Jupyter notebooks	analysis/, notebooks/, survey/	IMPLEMENTED
Real survey data, ethics approval	survey/survey_data.csv is synthetic	MISSING
Public-dataset benchmark results	code path exists; needs creditcard.csv in data/external	PENDING

Verification state when this pack was generated

Check	State
Unit tests (pytest)	VERIFIED

Check	State
Full pipeline run produced results	VERIFIED
Notebooks executed with stored outputs	VERIFIED
Docker Compose stack started and dashboards viewed	NOT VERIFIED
Four review decks generated	VERIFIED

Where a state says NOT VERIFIED the code exists but was not executed end-to-end in the environment used to prepare this pack. Details are in the validation report.

Repository map

Folder	Purpose
analysis/	7 focused analysis notebooks
notebooks/	01-08 research sequence
survey/	questionnaire, synthetic data, analysis notebook
src/	pipeline code by stage
models/	serialized models, evaluation, metadata (created by pipeline)
database/	schema, migrations, seed data
dashboard/	Grafana + Prometheus configuration
infrastructure/	Dockerfiles, Loki/Promtail config, helper scripts
config/	settings.yaml, logging.yaml
documentation/	9 documentation files
report/	6 chapter drafts, references.bib, figures, tables
review/	four monthly decks + generation scripts
tests/	pytest suite